

# Brain MRI Anomaly detection

An approach based on variational autoencoder (VAE)

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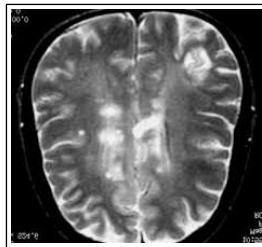
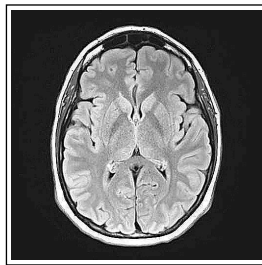
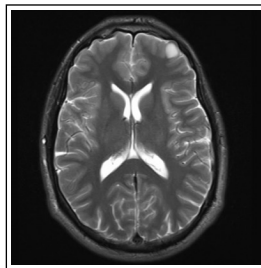
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# **1 Dataset & Methodology**

# Dataset Overview & Visualisation

While this dataset is originally designed for multi-class classification (categorizing tumors into Glioma, Meningioma, Pituitary, or No Tumor), it can also be adapted for unsupervised anomaly detection.

- **Source:** Brain Tumor MRI Dataset sourced from Kaggle
- **Content:** Multi-plane brain slices.
- **Preprocessing:**
  - Resizing to  $224 \times 224$
  - Intensity normalization  $[0, 1]$
  - Random Horizontal Flip



## Dataset Size

Healthy Samples *Used for training and baseline validation*

| Split Type           | Count        |
|----------------------|--------------|
| Training (80%)       | 1,116        |
| Validation (20%)     | 319          |
| Testing (10%)        | 160          |
| <b>Total Healthy</b> | <b>1,595</b> |



Anomalous Samples (Pathological) *Used for thresholding and final testing*

| Split Type             | Count      |
|------------------------|------------|
| Threshold Calib. (20%) | 181        |
| Final Testing (80%)    | 725        |
| <b>Total Anomalies</b> | <b>906</b> |

# VAE Logic for Anomaly Detection



Variational Autoencoders are generative models trained exclusively on images of healthy brain scans.

The VAE learns to compress these images into a lower dimensional latent space and then reconstruct them.

- **Assumption:** VAEs can only reconstruct data similar to the training set (Healthy).
- **Process:** Input  $x$  is encoded to  $z$  and decoded to  $\hat{x}$ .
- **Logic:**  $x - \hat{x} \approx 0$  for healthy tissue.
- **Score:** High Reconstruction Loss  $\Rightarrow$  Anomaly detected.

# Model Architecture



## 1. ENCODER

- **Layers:** 5 Strided Convolutions (reducing  $224 \times 224$  inputs down to  $7 \times 7$ ).
- **Features:** Integrated ResBlock modules at 64 and 128 channels to preserve gradient flow.
- **Activations:** Batch Normalization + LeakyReLU (0.2).

## 2. LATENT BOTTLENECK

- **Dense Layers:** Flattened to a 4,096-dimensional hidden layer with 10% Dropout.
- **Latent Space:** Maps to  $\mu$  and  $\log(\sigma^2)$  with a high-capacity Latent Dim of 2,048.

## 3. DECODER

- **Layers:** 5 Transpose Convolutions for symmetric upsampling back to  $224 \times 224$ .
- **Features:** Symmetrical ResBlock modules at 128 and 64 channels.
- **Output:** Sigmoid activation to match the normalized  $[0, 1]$  pixel range.

## 4. WEIGHT INITIALIZATION

- **Hidden Layers:** Kaiming Normal (optimized for LeakyReLU).
- **Linear/Output:** Xavier Normal.

# Architecture Diagram

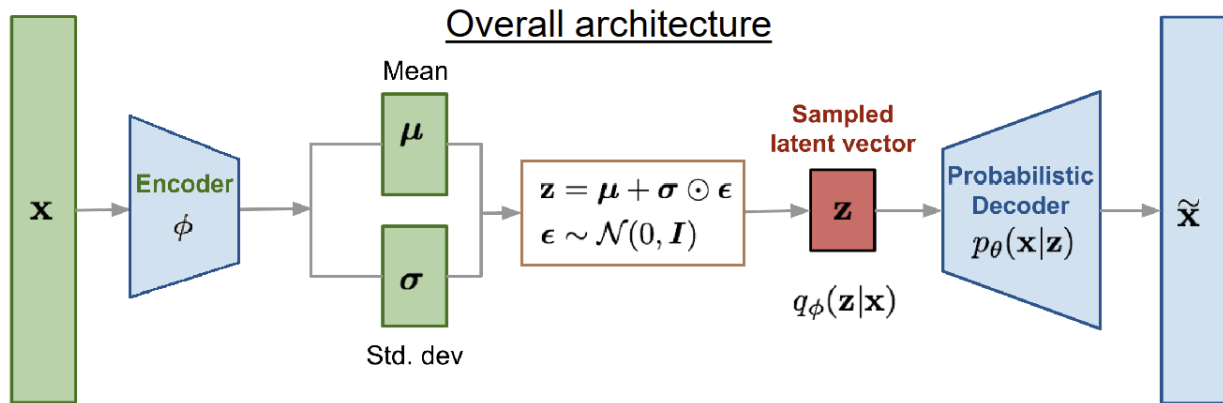


Figure 1: Visual Representation of the VAE Architecture

# The Composite Loss Function



To balance pixel-wise accuracy, structural fidelity, and latent regularization, the total loss is defined as:

$$\mathcal{L}_{\text{Total}} = \mathcal{L}_{\text{MaskedBCE}} + 5 \cdot \mathcal{L}_{\text{SSIM}} + \beta \cdot \mathcal{L}_{\text{KLD}}$$

## 1. Masked BCE ( $\mathcal{L}_{\text{MaskedBCE}}$ )

- Background Masking: Filters input using  $x > 0.05$ .

## 2. Structural Similarity ( $\mathcal{L}_{\text{SSIM}}$ )

- Evaluates the preservation of luminance, contrast, and structural textures.
- Weight ( $5 \times$ ): Manually scaled to match the BCE magnitude.

## 3. KL Divergence ( $\mathcal{L}_{\text{KLD}}$ )

- Regularization: Forces the latent space to approximate a standard normal distribution  $\mathcal{N}(0, I)$ .

Beta Annealing (Warmup): To prevent “posterior collapse”:

$$\beta(t) = \min \left( \beta_{\text{target}}, \beta_{\text{target}} \cdot \frac{t}{T_{\text{warmup}}} \right)$$

# Training Setup & Loss Curve

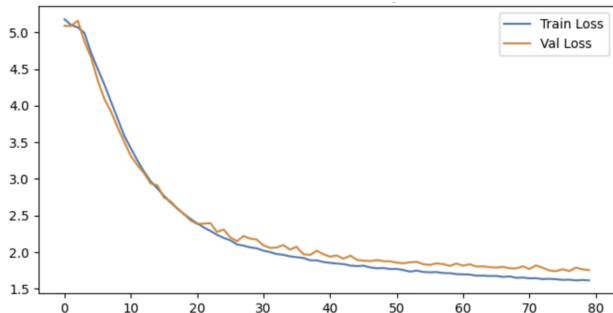


Figure 2: Training and Validation Loss

x-axis: epochs

y-axis: loss values

## Hyperparameters

| Parameter    | Value                      |
|--------------|----------------------------|
| Optimizer    | AdamW ( $\eta = 10^{-4}$ ) |
| Weight Decay | $10^{-5}$                  |
| Scheduler    | ReduceLROnPlateau          |
| Batch Size   | 32                         |
| Epochs       | 80 (Early Stop)            |
| Latent Dim   | 2048                       |

## 2 Results & Analysis

# Performance Metrics

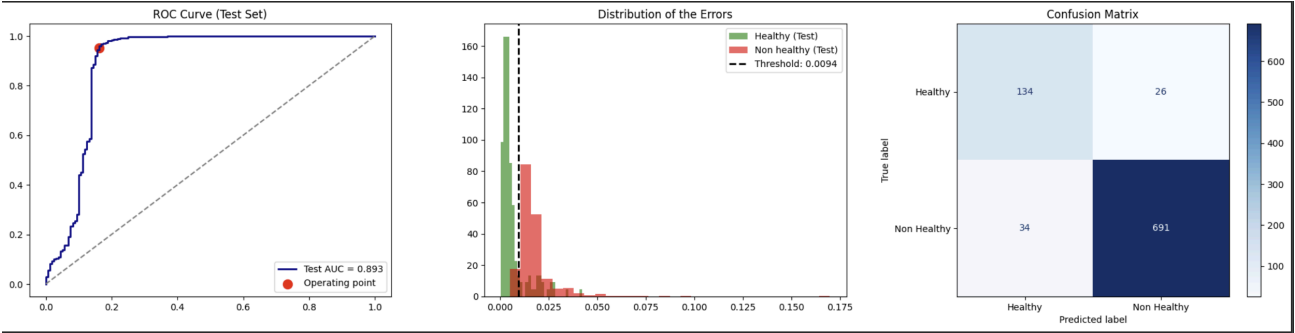


Figure 3: ROC Curve, Error Distribution, and Confusion Matrix

| Accuracy | AUC Score | Recall | Specificity | F1-Score | No Info Rate |
|----------|-----------|--------|-------------|----------|--------------|
| 0.9322   | 0.8926    | 0.9531 | 0.8375      | 0.9584   | 0.8192       |

# Anomaly Localization Pipeline

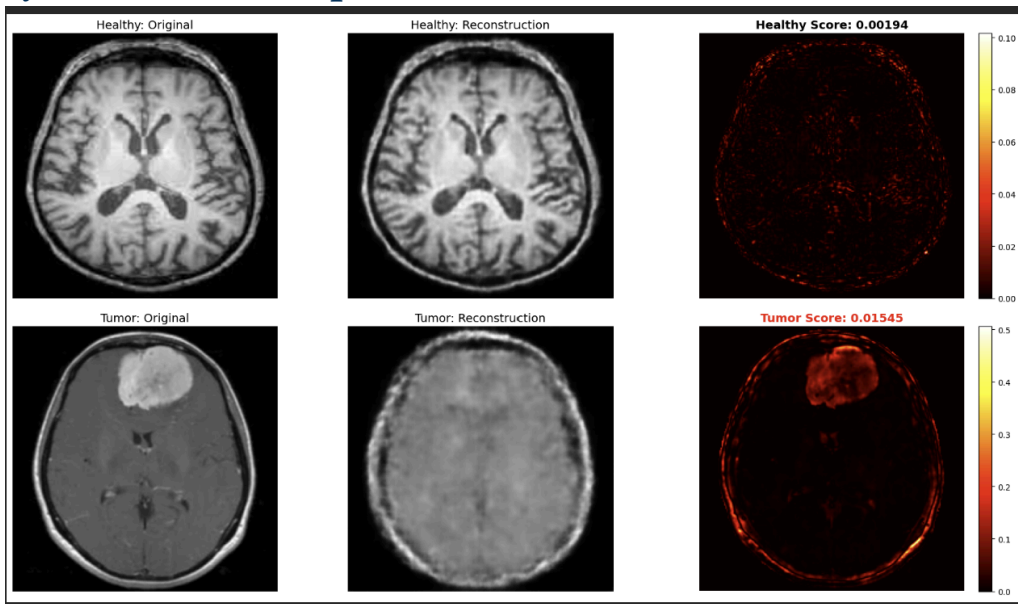


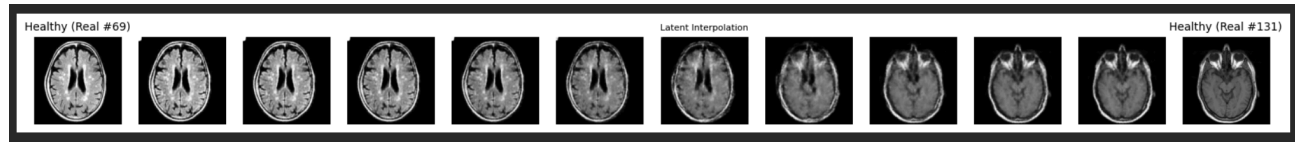
Figure 4: Visual Pipeline: Input ( $x$ ), VAE Reconstruction ( $\hat{x}$ ), and Heatmap Residual ( $|x - \hat{x}|$ )

# Latent Space Interpolation



By interpolating between two points  $z_1$  and  $z_2$  in the latent space:

- $z_{\text{interp}} = (1 - \alpha)z_1 + \alpha z_2$
- The reconstructions show smooth morphological transitions.
- This confirms that the model has learned a continuous representation of brain anatomy.



# Lighter Architecture (L-VAE)



To reduce model complexity, the following modifications were applied:

## 1. Architectural Changes

- Reduced parameters
- Bottleneck Restriction: Completely removed the redundant hidden dense layers connecting the encoder and decoder.
- Latent Compression: The latent dimension was drastically reduced from 2048 to 1024.

## 2. Loss Function Optimizations

- Transitioned from BCE to Masked L1 Loss.
- SSIM Rebalancing: Removed the 5x multiplier. Because L1 operates on a different numerical scale than BCE.

# Performance Comparison



| Metric               | Standard VAE<br>(140M) | Light VAE (40M) | Difference     |
|----------------------|------------------------|-----------------|----------------|
| Accuracy             | 0.9322                 | 0.9232          | -0.0090        |
| AUC Score            | 0.8926                 | 0.8630          | -0.0296        |
| Recall (Sensitivity) | 0.9531                 | 0.9655          | <b>+0.0124</b> |
| Specificity          | 0.8375                 | 0.7312          | -0.1063        |
| F1-Score             | 0.9584                 | 0.9537          | -0.0047        |

## 3 Wrap-up

# Limitations of the Approach



Despite architectural modifications, the models exhibit fundamental limitations stemming from the dataset's unstructured, multi-planar nature:

- **Latent Space Bias & Fragility:** The network prioritizes dominant data (Axial planes). Minority views (Coronal/Sagittal) are pushed to the latent edges, producing blurry statistical averages. When an anomaly like a tumor is introduced, it disrupts the entire structural mapping rather than causing a localized error.
- **Multi-Planar Geometric Variance:** Forcing a single, continuous latent space to simultaneously resolve vastly different macro-geometries introduces extreme spatial complexity into the training pipeline.

**Fundamental Takeaway:** Standard unconditional VAEs are fundamentally not equipped to model unstructured, multi-planar MRI data. This restricts the architecture's utility to broader image classification rather than precise spatial localization.

# Future Work



To overcome the limitations of unstructured data and latent space fragility, future iterations must focus on dataset standardization and conditional architectures.

## 1. Dataset Standardization

- Data Homogenization: Pre-processing the dataset to isolate single-plane views (e.g., exclusively Axial slices).
- Skull Stripping: Removing non-brain tissues.
- Specialized Ensembles: Training distinct models for each slice type.

## 2. Architectural Evolution: cVAE

- Conditional VAEs: Upgrading the network to a Conditional Variational Autoencoder (cVAE), **supported by a larger, highly structured dataset.**